

Label-Free Distribution Shift Detection in Truck Mode Classification via Uncertainty-Gap Monitors

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Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

Machine learning models in vehicle telematics face distribution shifts, but existing sequential monitors often require real-time ground-truth labels. We introduce the Uncertainty-Gap Monitor (UGM), a label-free empirical monitor based on the width of Venn-Abers probability intervals. In truck accelerometer data, UGM remained stable on benign shifts and produced large e-values that triggered alarms on the aggregated out-of-distribution fleet shifts.

Keywords: Conformal Prediction, Venn-Abers Predictors, Distribution Shift, Time Series

1. Introduction and Methodology

Industrial vehicle telematics face distribution shifts from mechanical degradation or novel sensor mounts. While Venn-Abers Testing of Exchangeability (VATE) flags such changes, it requires real-time true labels to accumulate e-values (Noureddinov, 2025). Since real-time labels are unfeasible in deployment, we introduce the Uncertainty-Gap Monitor (UGM), a label-free empirical score monitor.

A multi-class Inductive Venn-Abers Predictor (IVAP) (Vovk et al., 2015; Manokhin, 2017) utilizing a One-vs-Rest architecture calibrates base predictions for window x_n into a probability interval (p_0, p_1) . Operating parallel monitors for each class, we define a class’s empirical uncertainty score as its individual interval width (rather than an aggregate or solely the predicted class): $\Delta_n = p_1(x_n) - p_0(x_n)$. Intuitively, when the base classifier encounters unfamiliar feature spaces, its predictions become less reliable. The IVAP often reflects this increased uncertainty by outputting a wider probability interval, so we use Δ_n as a practical label-free shift score.

To handle discrete ties from IVAP’s isotonic regression, we compute a smoothed conformal p -value (Vovk et al., 2022) by comparing the streaming gap Δ_n against a multiset of N gaps from a held-out calibration set. Utilizing an independent tie-breaking variable $\tau_n \sim \text{Uniform}(0, 1)$:

$$p_n = \frac{|\{i : \Delta_i > \Delta_n\}| + \tau_n(|\{i : \Delta_i = \Delta_n\}| + 1)}{N + 1} \quad (1)$$

Wide intervals push p_n toward zero, which we transform using a power betting function (Vovk et al., 2021), $e_n = \epsilon p_n^{\epsilon-1}$. We then multiply these values into a running statistic, $M_n = M_{n-1}e_n$, and declare an alarm when M_n exceeds a preset threshold C . Because the score is compared against a fixed calibration set under serial dependence, we treat M_n as an empirical monitoring statistic rather than a formally valid martingale.

2. Empirical Investigation

We evaluated UGM on tri-axial accelerometer streams from 6 industrial trucks. A CatBoost classifier predicts operational modes (loading, parking, road) using engineered time-domain and frequency-domain features (e.g., RMS, spectral energy) extracted from 2-second sliding windows (26 Hz, 1-second stride) to capture the vibrational signature of each state. Because the windows overlap, the sequence is not exactly exchangeable, so we interpret UGM as a practical monitoring statistic. We set risk parameter $\epsilon = 0.5$ and alarm threshold $C = 10^5$, locking these default values before evaluating the test trucks.

Continuous data streams from Trucks 1 and 2 (primary fleet) were chronologically split into sets for base model training (258,080 windows), validation (128,865 windows), and IVAP calibration ($N = 86,039$ gaps). Without a temporal drop-buffer, a negligible 1-second shared-data overlap remains at the exact split boundaries due to the sliding stride. Truck 3 (primary fleet) served as an unseen subject-shift evaluation (85,620 windows). Across the calibration and subject-shift domains, Δ remained narrow. The accumulator M_n stayed safely below C without false alarms, reaching a terminal value of $< 10^{-300}$ due to expected numerical underflow from multiplicative decay.

Conversely, Trucks 4–6 from an independent fleet formed the aggregated out-of-distribution (OOD) domain-shift set (233,485 windows). Because this fleet introduces uncalibrated variations in suspension mechanics and sensor mounting angles, the uncertainty proxy increased. This widened Δ caused M_n to cross the alarm threshold C . In a retrospective Oracle Validation (Table 1), this unsupervised threshold crossing coincided with an 11.0-percentage-point drop in supervised accuracy (from 96.1% on the calibration set to 85.1%).

Table 1: Oracle Validation: UGM suppresses false alarms on benign variations while alerting on harmful domain shifts (Δ is macro-averaged across all class monitors).

Metric	Calibration	Subject shift (Truck 3)	OOD (Trucks 4-6)
Avg. Width (Δ)	0.0009	0.0007	0.0027
Final Statistic (M_n)	$< 10^{-300}$	$< 10^{-300}$	$> 10^{300}$
Oracle Accuracy	96.1%	96.6%	85.1%

While UGM successfully flags these mechanical changes, it fundamentally assumes domain shifts manifest as increased epistemic uncertainty (a widened gap). OOD shifts that induce narrow, overconfident predictions may go undetected. Future work will explore time-aware blocking and conditional conformal test martingales to restore exact theoretical validity under serial dependence.

Acknowledgments

This research was developed within the framework of the SEFLI project, funded by the German Federal Ministry for Economic Affairs and Energy (BMWE) under the Central Innovation Programme for SMEs (ZIM). We also extend our gratitude to KONVOI GmbH for their collaboration and provision of the operational truck dataset.

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